

altairsim — Quick Reference

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Getting out, and back in

Key	Does
<code>^E</code>	STOP — stop the machine and take the keyboard back. Nothing is lost.
<code>RUN</code>	Resume, at the exact instruction it stopped on.
<code>QUIT</code>	Leave. (There is no <code>EXIT</code> .)

Editing the command line

`Tab` completes what you are typing — a command, then a board id, then its property names, then a property's values (`SET mem0 fill=` then `Tab`); a second `Tab` lists the choices. `Up` / `Down` walk the command history, saved per directory in `.altairsim_history`.

Key	Does
<code>Ctrl-A</code> / <code>Ctrl-E</code> (or <code>Home</code> / <code>End</code>)	start / end of line
<code>Alt-B</code> / <code>Alt-F</code> (or <code>Ctrl-Left</code> / <code>Ctrl-Right</code>)	back / forward one word
<code>Ctrl-W</code> / <code>Ctrl-K</code> / <code>Ctrl-U</code>	erase word behind / to end of line / whole line
<code>Backspace</code> / <code>Delete</code>	erase before / under the cursor

Command line

```
altairsim [machine] [options]
```

machine	a built-in name, or a config file (has a '/' or ends .toml). Omitted: ./altairsim.toml if there is one, else `default`.
-m, --machine <n>	ALWAYS a built-in name -- never a file.
-f, --file <path>	ALWAYS a file -- never a built-in name.
-n, --none	empty backplane: no boards, no memory, nothing.
-l, --list	list the built-in machines and exit.
-s, --script <f>	run a command script, then exit with its status. Paths in it are relative to the script's folder.
-x, --exec <cmd>	run one monitor command (repeatable), then exit.
-i, --interactive	after --script/--exec, stay in the monitor.
--mcp	MCP server on stdio.
--mirror <sink>	with --mcp: mirror the console so a person can watch and take over. socket:PORT (telnet in), or on macOS and Linux pty or pty:LINK (open the link with a terminal program). Add ?ro for watch-only.
--log <file>	with --mcp: write what the guest prints to a text file, as it prints it. An ordinary file; starts empty each run.
-v, --version	print the version and exit.
-h, --help	print this help and exit.

Monitor commands

Type the part before the bracket.

Command	Does	Usage
B0[ARDS]	List, add, or remove boards on the backplane.	BOARDS [LIST] ADD <type> <id> [k=v...] REMOVE <id>
B[REAK]	Set a breakpoint on an address, memory/I/O access, or tape stop.	BREAK [<addr> MEM R W <addr> IO R W <port> TAPE STOP] [IF <expr> LOADS <expr>] [TRACE ON OFF]
COM[PARE]	Compare a range of memory against another address.	COMPARE <range> <addr>
C[ONFIG]	Load or save the whole machine as a TOML file.	CONFIG LOAD <f.toml> CONFIG SAVE <f.toml>
CONN[ECT]	Attach a serial unit to an endpoint (console, socket, file, ...).	CONNECT <id>:<u> <endpoint>
CONS[OLE]	Show or set the host console's properties.	CONSOLE [<k>=<v>...]
DE[POSIT]	Write bytes into memory at an address.	DEPOSIT <addr> <bytes...>
DI[SASM]	Disassemble memory into instructions.	DISASM [<addr> <range>] [n] [CPU=8080]
DISC[ONNECT]	Unplug the endpoint from a serial unit.	DISCONNECT <id>:<u>
D0	Run a file of monitor commands, one per line, as if typed.	D0 <file>
D[UMP]	Show memory as hex and ASCII.	DUMP [<addr> <range>] [WIDTH=16]
E[DIT]	Enter bytes into memory interactively from an address.	EDIT <addr> [ROM]
EX[AMINE]	Point the front panel at an address (and show that byte).	EXAMINE [<addr>]
F[ILL]	Fill a range of memory with a byte.	FILL <range> <byte>
HE[LP]	Show help for a command.	HELP [<command>]
H[ISTORY]	Replay the recent instruction (or bus-cycle) history.	HISTORY [BUS CPU] [n]
I[N]	Read a byte from an I/O port.	IN <port>

L[OAD]	Load a file into memory (binary, Intel hex or S-record).	LOAD <file> [AT <addr>] [FORMAT=BIN HEX SREC] [ROM]
MA[CHINE]	Load a built-in machine by name (MACHINE none empties the backplane).	MACHINE <name> MACHINE none
M[OUNT]	Put a disk or tape image into a drive; a .imd is converted to a raw .dsk beside it.	MOUNT <id>[:<u>] <file> [WP] [CREATE] [extract[=<base>]] [k=v...]
MOV[E]	Copy a range of memory to another address.	MOVE <range> <dest> [ROM]
N[EXT]	Step one instruction, running any CALL/RST to completion.	NEXT
NO[BREAK]	Remove a breakpoint, or all of them.	NOBREAK [id]
NOP[ASTE]	Stop a PASTE that is not finished.	NOPASTE
O[UT]	Write a byte to an I/O port.	OUT <port> <byte>

PA[STE]	Send a host file of any size to the guest's keyboard, as if pasted.	PASTE <file>
P[OWER]	Power-cycle the machine -- the only thing that clears RAM.	POWER
Q[UIT]	Leave the simulator.	QUIT
REGI[ON]	Add a RAM or ROM region to a memory board.	REGION ADD <id> type=ram rom at=<addr> [size= mount=]
RE[GS]	Show the CPU registers (SET REG changes one).	REGS SET REG <r>=<v>
RES[ET]	Reset the machine, keeping RAM (RESET CPU resets just the processor).	RESET [CPU]
REST[ORE]	Load machine state back from a snapshot.	RESTORE <file>
R[UN]	Start or resume the machine, optionally at an address.	RUN [addr]

SA[VE]	Write a range of memory out to a file.	SAVE <file> <range> [FORMAT=BIN HEX OCTAL PRN]
SEA[RCH]	Find bytes or a string in a range of memory.	SEARCH <range> <bytes...> "str"
SE[T]	Change a property of a board, the console, display, a register, the bus, or the machine.	SET <id>[:<u>] CONSOLE DISPLAY TERMINAL MACHINE REG BUS <k>=<v>
SH[OW]	Display the state of a board, the bus, or the machine.	SHOW <id> BOARDS BOARD <type> [UNITS] MACHINES MACHINE [<name>] BUS [MAP IO IRQ CONTENTION] ROMS MOUNTS PATHS CONSOLE DISPLAY SYMBOLS CLOCK VERSION
SN[APSHOT]	Save the whole machine state to a file.	SNAPSHOT <file>
STA[RTUP]	Edit the machine's boot list (the commands CONFIG SAVE saves as startup = [...]).	STARTUP [ADD <command> REMOVE <n> CLEAR]
S[TEP]	Run one instruction (or n), showing the registers after each.	STEP [n]

<code>SY[MBOLS]</code>	Load or clear a symbol table for disassembly.	<code>SYMBOLS LOAD <file> [REPLACE] SYMBOLS CLEAR</code>
<code>T[RACE]</code>	Log every bus cycle while the machine runs.	<code>TRACE ON OFF [file]</code> <code>[MASK=IN,OUT,IRQ,DMA,CONTENTION]</code>
<code>TY[PE]</code>	Feed text to the guest as if typed at its keyboard.	<code>TYPE "text"</code>
<code>U[NMOUNT]</code>	Take a disk or tape out of a drive.	<code>UNMOUNT <id>:<u></code>
<code>W[H0]</code>	Say which board answers an address or I/O port.	<code>WHO <addr> WHO IO <port></code>

Boards

CPU

Type	What it is
<code>8080</code>	MIT's 88-CPU: 8080A CPU board
<code>8085</code>	Generic 8085 CPU board
<code>z80</code>	Generic Z80 CPU board

Memory

Type	What it is
<code>bankmem</code>	Bank-switched RAM (Vector, Cromemco, North Star, ExpandoRAM)
<code>memory</code>	RAM/ROM board: plain, unbanked memory regions
<code>v2z80rom</code>	S100Computers V2 Z80: paged monitor EEPROM

Disk

Type	What it is
16fdc	Cromemco 16FDC: floppy controller + console UART, RDOS 2.52
64fdc	Cromemco 64FDC: floppy controller + console UART, RDOS 3.12
dcdd	MIT 88-DCDD: 8" hard-sector floppy controller
dualide	S100Computers IDE-AB: two CompactFlash sockets for CP/M 3
dualsd	S100Computers Dual SD: two microSD sockets for CP/M 3
fdcplus	FarmTek FDC+: serial drive (drive types 6, 7)
hdisk	MIT 88-HDSK Datakeeper: Pertec hard disk controller
icom	iCOM FD3712/FD3812: 8" floppy controller with boot PROM
mds	MIT 88-MDS: 5.25" minidisk controller
mdsa	North Star MDS-A: single-density 5.25" floppy controller
mdsad	North Star MDS-A-D: double-density 5.25" floppy controller
tarbell	Tarbell #1011: single-density floppy controller
tarbelldd	Tarbell #2022: double-density floppy controller
versafloppy	SD Systems VersaFloppy I/II: WD177x floppy controller

Serial

Type	What it is
2sio	MIT 88-2SIO: two 6850 serial ports
gsio	Generic SIO: two strap-configurable serial channels
io4	SSM IO-4: two serial + two parallel ports
pmmi	PMMI MM-103: Bell 103 modem
propio	S100Computers Console IO: Propeller console serial port
sbc	SD Systems SBC-100/200: Z80 single-board computer
sio	MIT 88-SIO: one serial port (COM2502 UART)
turnkey	MIT 8800b Turnkey Module: boot PROM, serial, auto-start

Tape

Type	What it is
acr	MTS 88-ACR: audio cassette interface
uio	MTS 88-UIO: serial port + cassette interface

Parallel and printer

Type	What it is
4pio	MTS 88-4PIO: up to four 6820 parallel ports
c700	MTS 88-C700: Centronics line-printer controller
d7a	Cromemco D+7A: analog + parallel I/O, joysticks
lpc	MTS 88-LPC: 88-LP line-printer controller
music6	Newtech Model 6 Music Board: 6-bit D/A and speaker
pio	MTS 88-PIO: 8-bit parallel port

Video

Type	What it is
cadzilla	CADzilla: HD63484 ACRTC graphics board with a Bt453 RAMDAC
dazzler	Cromemco Dazzler: color graphics
dazzler2	Cromemco Dazzler II: color graphics, on-board RAM
vdb8024	SD Systems VDB-8024: 80x24 video terminal board
vdm1	Processor Technology VDM-1: 16x64 memory-mapped video

Systems

Type	What it is
sol	Processor Technology Sol-PC I/O: serial, keyboard, tape

PROM programmer

Type	What it is
pb1	SSM PB1: 2708/2716 EPROM programmer

Other

Type	What it is
fp	Altair front panel: the SENSE switches at IN 0FFH
hostbridge	Host Bridge: file transfer to the host (not a period board)
rtc100	SciTronics RTC-100: battery-backed clock/calendar
ss1	CompuPro System Support 1: interrupts, timer, clock, serial
virtc	MITS 88-VI/RTC: vectored interrupts + real-time clock

Machines

Machine	What it is
8085	A minimal 8085 machine: an 8085 CPU, 64K of RAM, and a 2SIO console.
acuter	ACUTER at F000 -- CUTER on a plain Altair, with a terminal instead of a VDM.
altmon	An Altair with a monitor in ROM and a terminal on it.
amon	AMON 3.1 in a 4K EPROM at F000 -- Martin Eberhard's full-featured Altair monitor.
bankmem	A bank-switched RAM machine: a Z80, a console, and a Vector Graphic 64K bankmem.
basic4k	The machine Altair 4K BASIC was sold to run on: an 88-SIO Teletype, a cassette in the ACR.
basic8k	The machine Altair 8K BASIC was sold to run on: an 88-2SIO terminal, a cassette in the ACR.
cadzilla	An HD63484 ACRTC graphics board with a Bt453 RAMDAC in an Altair -- the bench for CADzilla.
cdbl	The default machine with the Combo Disk Boot Loader in the PROM socket.
compupro	A stock Altair with a CompuPro System Support 1 board for its clock/calendar.
cuter	CUTER 1.3 driving a Processor Technology VDM-1 -- the real Sol/CUTS monitor.
dazzler	A Cromemco Dazzler in an Altair -- the bench for the S-100's first color graphics card.
default	The machine you get when you name none: 56K, and the DBL boot PROM at FF00.
dualide	S100Computers "IDE-AB CF" machine -- a V2 Z80 CPU board booting CP/M 3 off a CompactFlash card.
dualidesd	S100Computers "IDE-AB CF+ESP32" machine -- both boards, CP/M 3 on CF drives A:/B: and SD drives C:/D:.
dualsd	S100Computers "Dual SD" machine -- a V2 Z80 CPU board booting CP/M 3 off a microSD card.
icom	iCOM FD3712 8" floppy machine -- boots CP/M 2.2 off a single-density iCOM disk.
lineprinter-lpc	The default machine with an 88-LPC line printer at port 02; CONNECT <code>lpt0:prn</code> to a file or the console.
lineprinter	The default machine with an 88-C700 line printer at port 02; CONNECT <code>lpt0:prn</code> to a file or the console.
minidisk	The Altair Minidisk: an 88-MDS at 08 and the MDBL boot PROM. You supply the 5.25" disk.
northstar	North Star MDS-A single-density floppy: the controller at E800 and its boot PROM. You supply the disk.
northstardd	North Star MDS-A-D double-density floppy: the controller at E800 and its boot PROM. You supply the disk.
original	The Altair as it actually left Albuquerque.

parallel	The default machine with two MITS parallel boards: an 88-PIO and an 88-4PIO.
ps2	The machine MITS Programming System II ran on: basic8k's cards, but not basic8k's bootstrap.
ps2int	MITS Programming System II, WITH INTERRUPTS. ps2 with A9 down and an 88-VI/RTC in it.
rombasic	MITS Extended ROM BASIC 16K -- Extended BASIC that runs directly out of ROM.
sbc200	SD Systems SBC-200 -- a 4 MHz Z80 single-board computer running the MSMONR21 monitor.
sbc200v	The SD Systems SBC-200 with a VDB-8024 VIDEO console: SDMONV21 on an 80x24 screen.
sol20	The Processor Technology Sol-20 -- an integrated 8080 machine, running SOLOS.
tarbell	Tarbell #1011 single-density floppy machine -- boots CP/M 2.2 the moment a disk is in it.
tarbelldd	Tarbell #2022 double-density floppy machine -- boots CP/M 2.2 the moment a disk is in it.
turnkey	The MITS 8800bt -- an Altair with a Turnkey Module where the front panel used to be.
vdm1	A Processor Technology VDM-1 in an Altair, and a demo that draws on it.
z80	A minimal Z80 machine: a z80 CPU, 64K of RAM, and a 2SIO console.

A machine file, in one look

```
[machine]
name      = "mine"
base      = "default"      # start from a machine, and say what is DIFFERENT
startup   = ["RUN FF00"]    # the operator's own keystrokes. There is no BOOT verb.

[[board]]
# type + a NEW id      -> ADD the card
type = "2sio"          # type + an id from the base -> REPLACE it outright
id   = "sio0"           # NO type + an id      -> MODIFY the one already there
port = 10               # remove = true        -> PULL THE CARD OUT

[board.unit.a]          # a unit's own settings
connect = "console"

[[board.region]]        # a list the card owns (memory)
type = "ram"
at   = 0000              # hex: it is an address
size = "56K"             # decimal: it is a size

[[board.drive]]         # a list the card owns (disk controllers)
unit = 0
mount = "cpm.dsk"        # relative to THIS FILE

[console]               # the HOST's terminal -- not a board
strip7out = true
base      = octal        # read/print the wire class in split octal (MITS style)
```

Paths: a path *inside* a machine file is relative to **that file**. A path you *type* is relative to **your shell**.

Endpoints — `CONNECT <id>:<unit> <endpoint>`

Endpoint	Is
<code>console</code>	the host terminal. Exactly one unit may hold it.
<code>null</code>	nowhere. Writes vanish, reads never come.
<code>loopback</code>	itself — what you write comes back.
<code>scripted</code>	a caller in place of a human — what MCP and the tests type into.
<code>socket:PORT</code>	listens , as a raw pipe — for a program at the far end. <code>?banner</code> greets each caller.
<code>socket:HOST:PORT</code>	calls out , as a raw pipe.
<code>telnet:PORT</code>	listens , speaking Telnet — this is telnet-in for a person. Greets each caller; <code>?banner=off</code> stops it.
<code>telnet:HOST:PORT</code>	calls out , taking the telnet client's part.
<code>serial:DEVICE</code>	a real serial port on this host.
<code>pty</code>	a pseudo-terminal (macOS and Linux): open <code>/tmp/altairsim0</code> with a terminal program. <code>pty:LINK</code> selects the name.
<code>in:PATH</code>	a host file as a reader (paper tape). <code>?cps=N</code> paces it.
<code>out:PATH</code>	a host file as a punch — 8-bit clean, never truncating.
<code>terminal</code>	a window the simulator draws itself (SDL builds). <code>?emulation=vt100 adm3a vt52 h19</code> , <code>?size=COLSxROWS</code> .
<code>printer:QUEUE</code>	a real print queue on this host.
<code><endpoint> FILE</code>	a tap: append <code> FILE</code> to any endpoint to also log the line. <code>?fmt=text</code> = plain text, only what the guest prints.
<code><endpoint> socket:PORT</code>	a live mirror: <code>telnet</code> in to watch and take over. <code>?ro</code> = watch-only.
<code><endpoint> pty[:LINK]</code>	the same mirror on a pseudo-terminal (macOS and Linux): open <code>/tmp/altairsim0</code> , or <code>LINK</code> , with a terminal program.
<code><endpoint> serial:DEVICE</code>	the same mirror on a real serial port, 8N1: a hardware terminal watches and types. <code>?baud=N</code> (default 9600), <code>?ro</code> .